SPLITTING WITH A SCHEME SOURCE WHEN S, T ARE FLAT AND FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers within this span. The section-variant input is declared external.

Allowed standard prerequisites (editorial).

- Splitting existence. lemma-splitting (carded L26659). Construction of U' = (R_s/U, e)_fin whose scheme nature is shown.
- Scheme nature of the sectioned finite part. lemma-finite-separated-flat-locally-finite-presentation (carded L26657) and lemma-finite-plus-section (same chapter, NOT carded, declared external). U' is a scheme separated over U.

Source reference. Original lemma 04RR: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2168–2168. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04RI: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2082–2082. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DTC: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2583–2583. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DTB: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2479–2479. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0DTB assume in addition that s, t are flat and locally of finite presentation. Then there exists a scheme U', a separated étale morphism $U' \to U$, and a point $u' \in U'$ lying over u with $\kappa(u) = \kappa(u')$ such that the restriction $R' = R|_{U'}$ of R to U' is split over u'.*

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Proof. This follows from the construction of U' in the proof of Lemma 0DTC because in this case $U' = (R_s/U, e)_{fin}$ is a scheme separated over U by Lemmas 04RI and 04RR. □